AI-Powered Trusted Vehicle Marketplace

Complete Project Documentation and Software Requirements Specification (SRS)

Buy & Sell | Rental | Trusted Deals | AI Recommendations

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PART A: PROJECT DOCUMENTATION

1. Introduction

1.1 Project Overview

The AI-Powered Trusted Vehicle Marketplace is a web and mobile platform where users can buy and sell second-hand cars and bikes, rent vehicles from owners, and make decisions with the help of AI. The platform's central promise is trust: verified sellers, checked documents, genuine photos, transparent scoring and secure bookings.

1.2 Objectives

- Provide one platform for used-vehicle trading and vehicle rental.
- Reduce fraud and fake listings through verification and a Vehicle Health Score.
- Help buyers judge prices with an AI fair-price estimate.
- Help owners decide whether to sell or rent using a Sell-or-Rent Calculator.
- Recommend suitable vehicles based on a user's needs, budget and usage.

1.3 Scope

In scope: listing management, search and filters, user and seller verification, Vehicle Health Score, Sell-or-Rent Calculator, rental booking requests, reviews, in-app messaging, payments or deposits, AI recommendations, admin panel.

Out of scope (initial release): vehicle financing or loans, doorstep inspection fleet, vehicle delivery logistics, insurance underwriting, and new-vehicle sales.

1.4 Target Users

- **Buyers:** individuals looking for a used car or bike.
- **Sellers/Owners:** individuals or small dealers listing vehicles.
- **Renters:** people needing a vehicle for days or weeks (tourists, students, commuters).
- **Administrators:** platform staff who verify and moderate.

2. Problem Statement

- Classified platforms suffer from fake listings, fraud and heavy bargaining.
- Buyers cannot easily tell whether a price is fair or a vehicle's condition is genuine.
- Owners do not know if selling or renting their vehicle is more profitable.
- Buy/sell and rental exist on separate apps, with no connection between the two journeys.
- Peer-to-peer rental lacks clear rules for damage, insurance and disputes.

3. Proposed Solution

Module	Description
Buy and Sell	Owners list used cars and bikes; buyers search by price, location, model and condition.
Rental	Owners offer vehicles for rent; customers request bookings by date, duration and location.
Trusted Deals	Seller verification, document checks (RC, insurance, PUC), genuine photos, reviews, secure booking.
Vehicle Health Score	A 0-100 score shown on every listing summarising trust and condition.
Sell-or-Rent Calculator	Tells an owner whether selling or renting is likely to earn more.
AI Recommendations	Needs-based vehicle suggestions, fair-price estimates, fraud detection.

4. Market and Competitor Analysis

Segment	Existing players	Gap we target
Used vehicle trading	OLX, Cars24, Spinny, Droom, CarDekho	Per-listing transparent trust score; owner decision tools.
Vehicle rental	Zoomcar, Revv, SwapRides, RaideX	Link between rental and buying (try-before-you-buy, future scope).

Note: competitor information is based on a preliminary review and should be re-verified before any investor or public use. Having both buy/sell and rental on one platform is not by itself a differentiator; the differentiators are trust, transparency and AI-assisted decisions.

5. Signature Features in Detail

5.1 Vehicle Health Score (0-100)

A composite score displayed on every listing. The weights below are an initial proposal and should be tuned with real data.

Component	Weight	Inputs
Document verification	30	RC validity, insurance validity, PUC, owner name match
Service and challan history	20	Uploaded service records, pending challans
Photo quality and authenticity	20	Guided in-app photos, duplicate or stock image detection
Seller verification	15	Phone, government ID, address check
Reviews and platform history	15	Ratings, completed deals, complaint history

Display bands: 85-100 Excellent, 70-84 Good, 50-69 Average, below 50 Needs caution. Listings below a minimum threshold may be hidden or flagged until improved.

5.2 Sell-or-Rent Calculator

The owner enters model, year, km driven, condition and city. The system estimates:

- **Sale value:** AI fair-price estimate (see Section 6).

- **Monthly net rental income** = (daily rate x expected rental days per month) - (maintenance + insurance share + platform fee + cleaning/other costs).
- **Break-even months** = estimated sale value / monthly net rental income.
- **Depreciation adjustment:** expected fall in sale value over the same period, so the owner sees the true cost of waiting.
- **Recommendation:** Sell, Rent, or Either, with the numbers shown.

All outputs are estimates and must be labelled as such. Rental income shown is subject to the legal eligibility of the vehicle for rental use.

6. AI Module Design

Capability	Approach	Data needed
Needs-based recommendation	User describes needs in natural language; an LLM extracts constraints (budget, distance, mileage, seats) and the system ranks matching listings, returning reasons.	Listings with structured attributes; user query
Fair-price estimate	Regression model (e.g., gradient boosting) using model, year, km, city, condition, owners count.	Historical listing and sale prices
Fraud and fake-listing detection	Image hashing for duplicates, reverse-match with stock photos, price anomaly rules, odometer inconsistency checks.	Uploaded images, listing history
Rental suggestion	Rule-based matching on trip type, group size, luggage and duration.	Rental inventory
Listing assistant	Generates description from guided photos and form inputs; flags missing angles.	Seller photos and inputs

Cold-start strategy: in the first months there will not be enough data for trained models. The system starts with rule-based logic and public reference price ranges, then moves to trained models as listings and completed transactions accumulate. AI suggestions are advisory; users make the final decision.

7. System Architecture and Technology Stack

A layered architecture with a mobile/web client, an API layer, core services and a data layer. The stack below is a suggestion and can be changed by the development team.

Layer	Components	Suggested technology
Client	Web app, mobile app (Android first)	React (Vite) with Tailwind CSS in JavaScript; mobile app later (React Native)
API	REST API, authentication, rate limiting	Node.js or Django / FastAPI
Core services	Listings, Search, Rental booking, Verification, Health Score, Calculator, Messaging, Notifications, Payments	Modular services or a modular monolith for the MVP
AI services	Price model, recommendation, fraud detection	Python (scikit-learn / XGBoost), LLM API
Data	Relational DB, cache, object storage, search index	PostgreSQL, Redis, S3-compatible storage, Elasticsearch or PostgreSQL full-text

Layer	Components	Suggested technology
Integrations	Payment gateway, SMS/OTP, maps, KYC and vehicle-data verification, email	Razorpay or similar, maps API, authorised verification providers
DevOps	Hosting, CI/CD, monitoring, backups	Cloud provider, Docker, GitHub Actions

MVP recommendation: start as a modular monolith to move fast, and split into separate services only when scale demands it.

8. Database Design

Main entities and their key fields (primary keys and standard timestamps omitted for brevity).

Entity	Key fields
User	name, phone, email, role, verification_status, city, created_at
SellerProfile	user_id, id_proof_ref, address_proof_ref, rating, deals_completed
Vehicle	owner_id, type (car/bike), make, model, year, km_driven, fuel, owners_count, reg_number, city, condition_notes
Listing	vehicle_id, listing_type (sale/rent), price, status, health_score, created_at
VehicleDocument	vehicle_id, doc_type (RC/insurance/PUC/service), file_ref, expiry_date, verified_by, status
VehiclePhoto	vehicle_id, angle, file_ref, hash, authenticity_flag
RentalOffer	listing_id, daily_rate, deposit, min_days, max_days, availability_calendar
Booking	offer_id, renter_id, start_date, end_date, amount, deposit, status
Payment	booking_or_listing_ref, payer_id, amount, type, gateway_ref, status
Review	reviewer_id, target_user_id, listing_or_booking_ref, rating, comment
Message	sender_id, receiver_id, listing_id, body, sent_at
HealthScoreLog	listing_id, component_scores, total_score, computed_at
CalculatorRun	user_id, vehicle_inputs, sale_estimate, rental_estimate, recommendation
Report/Dispute	reporter_id, target_ref, reason, status, resolution

Key relationships: User 1-N Vehicle; Vehicle 1-N Listing; Vehicle 1-N VehicleDocument and VehiclePhoto; Listing 1-1 RentalOffer (for rent listings); RentalOffer 1-N Booking; Booking 1-N Payment.

9. User Roles and Workflows

Role	Key permissions
Guest	Browse listings, view Health Score, use AI search (limited).
Buyer/Renter	Save listings, contact sellers, request bookings, pay, review.
Seller/Owner	Create and manage listings, upload documents and photos, respond to enquiries, use calculator, manage rental calendar.
Admin/Verifier	Verify users and documents, moderate listings, resolve disputes, view analytics.

9.1 Seller Workflow

- Register with phone OTP, complete ID verification.

- Add vehicle details; capture guided photos in-app; upload documents.
- Platform verifies documents and computes the Health Score.
- Optionally run the Sell-or-Rent Calculator.
- Publish listing for sale and/or rent; respond to enquiries.

9.2 Buyer Workflow

- Search or describe needs to the AI assistant.
- View ranked results with Health Score and fair-price indicator.
- Chat with seller, schedule inspection, finalise deal (payments or token amount through platform if enabled).

9.3 Rental Workflow

- Renter selects dates and location, and sends a booking request.
- Owner accepts or declines.
- Renter pays rent and refundable deposit; pickup photo check recorded.
- On return, photos are compared; deposit is released or adjusted; both sides leave reviews.

10. Security, Privacy and Legal Compliance

10.1 Security

- HTTPS everywhere; password hashing; OTP-based login; role-based access control.
- Encrypted storage of ID documents with restricted access and audit logs.
- Payments handled by a certified gateway; the platform does not store card data.
- Rate limiting, input validation and regular vulnerability testing.

10.2 Privacy

The platform will handle personal data (identity documents, phone numbers, vehicle registration details). It should be designed to comply with India's Digital Personal Data Protection Act, 2023 and the IT Act, 2000, including consent, purpose limitation, data minimisation, retention limits and a grievance process.

10.3 Legal and Regulatory Considerations

- **Rental:** self-drive vehicle rental in India generally requires commercial registration/permits under the Motor Vehicles Act and state rules. Renting a private-registered vehicle peer-to-peer is a legal grey area. Obtain legal advice before enabling rentals; consider launching with licensed owners or partners.
- **Sale:** transfer of ownership (RTO process, Form 29/30, NOC where required) must follow the law; the platform facilitates, but does not replace, the legal transfer.
- **Intermediary liability:** define the platform's role, terms of use, and listing policies; implement takedown and grievance procedures.
- **Insurance:** clarify coverage during rentals and partner with an insurer where possible.
- **Disclaimer:** this section is informational and not legal advice; confirm with a qualified lawyer.

11. Business Model

Revenue stream	How it works
Rental commission	Percentage of each completed booking.
Listing fees / featured listings	Optional paid boost or premium listing for sellers and dealers.
Verification / inspection fee	Paid document verification or inspection add-on.
Lead fees from dealers	Subscription for verified dealer accounts.
Partner revenue	Insurance, financing and service partner referrals (later phase).

Pricing should be validated through user interviews; individual sellers should be able to list free at launch to build supply.

12. Testing Strategy

Test type	Focus
Unit testing	Score computation, calculator formulas, validation logic.
Integration testing	API with database, payment gateway sandbox, verification providers.
System/functional testing	End-to-end seller, buyer and rental flows.
Security testing	Authentication, authorisation, data exposure, file upload.
Performance testing	Search latency and concurrent user load.
AI evaluation	Price-estimate error (MAPE), fraud-detection precision/recall, recommendation relevance via user feedback.
User acceptance testing	Pilot with 20-30 real users in the launch city.

13. Project Roadmap

Phase	Duration	Deliverables
Phase 0: Validation	Month 1	City and segment chosen; interviews with 20-30 users and owners; legal review; wireframes.
Phase 1: MVP	Month 2	Registration, listings, search, basic Health Score, contact/booking requests; 15-20 manually onboarded verified listings.
Phase 2: Beta	Month 3	Sell-or-Rent Calculator, reviews, messaging, payments sandbox; real-user testing and first paid transactions.
Phase 3: AI	Month 4-6	Fair-price model, needs-based recommendations, fraud detection, listing assistant.
Phase 4: Scale	Month 6+	More cities and segments, partnerships, mobile app polish, analytics.

Launch strategy: begin with one city and one segment (for example verified used bikes or local bike rental) and onboard both sides manually to solve the chicken-and-egg problem.

14. Risk Analysis

Risk	Impact	Mitigation
Rental legality and permits	High	Legal review; start with licensed partners; phase rental carefully.
Insurance and damage disputes	High	Deposit, photo check at pickup/return, insurance partner, clear terms.
Fraud and fake documents	High	Multi-step verification, AI checks, report and ban system.
Low initial liquidity (buyers/sellers)	High	Single city/segment, manual onboarding, local partnerships.
Strong incumbents	Medium	Differentiate on trust, transparency and owner tools.
Insufficient data for AI	Medium	Rule-based start; improve models as data grows.
Data breach or privacy violation	High	Encryption, access control, compliance with data-protection law.
Dependence on third-party verification APIs	Medium	Fallback manual verification; multiple providers.

15. Future Scope

- **Try-before-you-buy:** rent a vehicle for 1-2 days and adjust the rent against the purchase price.
- Vernacular and voice search (Hindi and regional languages).
- Total cost of ownership view (fuel, service, insurance, resale).
- Photo-based AI condition reports.
- Financing and insurance partnerships.
- Dealer dashboard and bulk listing tools.
- Expansion to more cities and vehicle types.

PART B: SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Prepared in the structure of IEEE 830 / ISO/IEC/IEEE 29148.

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements of the AI-Powered Trusted Vehicle Marketplace. It is intended for the development team, testers, project guides/stakeholders and future maintainers.

1.2 Scope

The system is a web and mobile application that lets users list, search, buy and sell used cars and bikes, rent vehicles, view a Vehicle Health Score, run a Sell-or-Rent Calculator, and receive AI-based recommendations. It also provides an administration panel for verification and moderation.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
RC	Registration Certificate of a vehicle
PUC	Pollution Under Control certificate
KYC	Know Your Customer identity verification
OTP	One-Time Password
Health Score	0-100 trust and condition score of a listing
MVP	Minimum Viable Product
RBAC	Role-Based Access Control
API	Application Programming Interface
Listing	A vehicle posted for sale or rent
Deposit	Refundable amount held during a rental

1.4 References

- IEEE 830-1998 / ISO/IEC/IEEE 29148 (requirements engineering).
- Motor Vehicles Act, 1988 and applicable state rules.
- Information Technology Act, 2000; Digital Personal Data Protection Act, 2023.

1.5 Document Overview

Section 2 gives an overall description; Section 3 lists functional requirements; Section 4 covers interfaces; Section 5 non-functional requirements; Section 6 use cases; Section 7 priorities and traceability.

2. Overall Description

2.1 Product Perspective

A new, standalone, client-server system with web and mobile clients, a backend API, a database, and third-party integrations (payments, SMS, maps, verification).

2.2 Product Functions (summary)

- User registration, login and verification
- Vehicle listing for sale and rent
- Search, filter and compare
- Vehicle Health Score
- Sell-or-Rent Calculator
- AI recommendations, price estimate, fraud checks
- Rental booking and deposit handling
- Reviews, ratings and messaging
- Admin verification, moderation and dispute handling

2.3 User Classes and Characteristics

User class	Characteristics
Buyer/Renter	General public; varied technical skill; mobile-first; needs simple UI.
Seller/Owner	Individuals or small dealers; need easy listing and photo upload.
Administrator/Verifier	Trained staff; need dashboards for review queues.

2.4 Operating Environment

- Web: current versions of Chrome, Edge, Firefox, Safari.
- Mobile: Android 8+ (first release); iOS later.
- Server: Linux-based cloud hosting.

2.5 Design and Implementation Constraints

- Compliance with Indian data-protection and IT laws.
- Rental features subject to legal clearance.
- Use of third-party APIs subject to their terms and availability.
- Limited training data for AI at launch.

2.6 Assumptions and Dependencies

- Users have internet access and a smartphone.
- Verification data providers are available and authorised for use.
- A payment gateway account is approved.
- Initial launch in a single city.

3. Functional Requirements

Priority key: M = Must have (MVP), S = Should have, C = Could have (later).

3.1 User Management (UM)

ID	Requirement	Priority
FR-UM-01	The system shall allow registration using a mobile number verified by OTP.	M
FR-UM-02	The system shall allow login, logout and secure session management.	M
FR-UM-03	The system shall let users create and edit a profile (name, city, photo).	M
FR-UM-04	The system shall let sellers/owners submit ID proof for verification.	M
FR-UM-05	The system shall show a verified badge on profiles that pass verification.	M
FR-UM-06	The system shall support role-based access (guest, user, seller, admin).	M
FR-UM-07	The system shall support password reset and account deletion on request.	S

3.2 Listing Management (LM)

ID	Requirement	Priority
FR-LM-01	The system shall allow a seller to create a listing for sale, for rent, or both.	M
FR-LM-02	The system shall capture vehicle type, make, model, year, km driven, fuel, owner count, registration number and city.	M
FR-LM-03	The system shall require minimum guided photos (front, rear, sides, interior, odometer) captured through the app.	M
FR-LM-04	The system shall allow upload of RC, insurance, PUC and service records.	M
FR-LM-05	The system shall let sellers edit, pause, mark sold or delete their listings.	M
FR-LM-06	The system shall hold new listings in a pending state until basic verification passes.	S
FR-LM-07	The system shall detect duplicate listings by registration number.	S

3.3 Search and Discovery (SD)

ID	Requirement	Priority
FR-SD-01	The system shall allow search by keyword and filters: price, location, make/model, year, km, fuel, type, condition.	M
FR-SD-02	The system shall allow sorting by price, date, distance and Health Score.	M
FR-SD-03	The system shall allow users to save listings and searches.	S
FR-SD-04	The system shall allow side-by-side comparison of up to 3 vehicles.	S
FR-SD-05	The system shall display listing details with photos, documents status and Health Score.	M

3.4 Vehicle Health Score (HS)

ID	Requirement	Priority
FR-HS-01	The system shall compute a 0-100 Health Score for each listing using weighted components (documents, service/challan, photos, seller verification, reviews).	M
FR-HS-02	The system shall display the score and a breakdown of its components on the listing page.	M
FR-HS-03	The system shall recompute the score when documents, photos or seller data change.	M
FR-HS-04	The system shall show suggestions to the seller on how to improve the score.	S
FR-HS-05	The system shall allow administrators to configure component weights.	S
FR-HS-06	The system shall log each score computation for audit.	S

3.5 Sell-or-Rent Calculator (SR)

ID	Requirement	Priority
FR-SR-01	The system shall accept vehicle details and optional cost inputs from the owner.	M
FR-SR-02	The system shall estimate the sale value of the vehicle.	M
FR-SR-03	The system shall estimate monthly net rental income and break-even months using the defined formula.	M
FR-SR-04	The system shall output a recommendation (Sell, Rent or Either) with supporting figures.	M
FR-SR-05	The system shall label all results as estimates and show the assumptions used.	M
FR-SR-06	The system shall let the owner create a sale or rental listing directly from the result.	S
FR-SR-07	The system shall warn when a vehicle may not be legally eligible for rental.	S

3.6 Rental Management (RM)

ID	Requirement	Priority
FR-RM-01	The owner shall be able to set daily rate, deposit, minimum/maximum days and availability calendar.	M
FR-RM-02	The renter shall be able to search available vehicles by date, duration and location.	M
FR-RM-03	The renter shall be able to send a booking request; the owner shall accept or decline within a set time.	M
FR-RM-04	The system shall prevent double booking of the same vehicle for overlapping dates.	M
FR-RM-05	The system shall collect rent and a refundable deposit through the payment gateway.	M
FR-RM-06	The system shall record pickup and return photos and timestamps for each booking.	S
FR-RM-07	The system shall release or adjust the deposit after return and dispute window.	M
FR-RM-08	The system shall verify the renter's driving licence before confirming a booking.	M
FR-RM-09	The system shall support cancellation with a defined refund policy.	S

3.7 AI Recommendation and Intelligence (AI)

ID	Requirement	Priority
FR-AI-01	The system shall accept a natural-language description of needs and return ranked suggestions with reasons.	S
FR-AI-02	The system shall show a fair-price indicator (below, fair, above market) on listings.	S
FR-AI-03	The system shall flag suspected fake listings (duplicate or stock photos, price anomalies) for admin review.	S
FR-AI-04	The system shall suggest rental vehicles based on trip type, group size and duration.	C
FR-AI-05	The system shall help sellers generate a listing description and detect missing photo angles.	C
FR-AI-06	The system shall fall back to rule-based suggestions when the AI service is unavailable or data is insufficient.	M
FR-AI-07	AI outputs shall be presented as suggestions, not guarantees.	M

3.8 Communication and Reviews (CR)

ID	Requirement	Priority
FR-CR-01	The system shall provide in-app messaging between buyer and seller, and renter and owner.	M
FR-CR-02	The system shall mask phone numbers until the user chooses to share them.	S
FR-CR-03	The system shall allow ratings and reviews only after a completed deal or booking.	M
FR-CR-04	The system shall allow users to report listings, users and messages.	M
FR-CR-05	The system shall send push, SMS or email notifications for key events (enquiry, booking, payment).	S

3.9 Payments (PM)

ID	Requirement	Priority
FR-PM-01	The system shall integrate a payment gateway for rent, deposits and optional token amounts.	M
FR-PM-02	The system shall generate receipts and maintain transaction history.	M
FR-PM-03	The system shall support refunds and payouts to owners after deducting the platform fee.	M
FR-PM-04	The system shall not store card or UPI credentials.	M

3.10 Administration (AD)

ID	Requirement	Priority
FR-AD-01	Admins shall review and approve or reject user and document verification.	M
FR-AD-02	Admins shall moderate, edit or remove listings and suspend users.	M
FR-AD-03	Admins shall view and resolve reports and disputes.	M
FR-AD-04	Admins shall view dashboards (users, listings, bookings, revenue).	S
FR-AD-05	The system shall keep an audit log of admin actions.	M

4. External Interface Requirements

4.1 User Interfaces

- Responsive web interface and Android app with a consistent design.
- Simple, mobile-first screens; support for English and Hindi (more languages later).
- Accessibility: readable fonts, sufficient contrast, clear error messages.

4.2 Hardware Interfaces

Device camera (guided photo capture) and GPS/location (nearby search), with user permission.

4.3 Software and Communication Interfaces

Interface	Purpose
Payment gateway API	Rent, deposits, refunds, payouts
SMS/OTP provider	Login and notifications
Maps and geocoding API	Location search and distance
KYC / vehicle verification provider	Identity and RC verification, subject to authorisation
Email and push service	Notifications
LLM / ML services	Recommendations and assistants
Protocols	HTTPS/TLS, REST over JSON, WebSocket for chat

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Search results shall load in under 3 seconds for 95% of requests under normal load; pages shall be usable on 4G networks.
NFR-02	Scalability	The system shall support at least 10,000 registered users and 500 concurrent users at launch, and be scalable horizontally.
NFR-03	Availability	Target uptime of 99.5% excluding planned maintenance.
NFR-04	Security	All traffic over HTTPS; passwords hashed; RBAC; protection against OWASP Top 10 risks; encrypted storage of identity documents.
NFR-05	Privacy	Compliance with DPDP Act 2023: user consent, data minimisation, deletion on request, retention policy.
NFR-06	Reliability	Daily database backups with tested restore; payment operations must be idempotent.
NFR-07	Usability	A new seller shall be able to publish a listing in under 10 minutes without training.
NFR-08	Maintainability	Modular code, documented APIs, automated tests with at least 70% coverage on core logic.
NFR-09	Compatibility	Support latest two versions of major browsers and Android 8 and above.
NFR-10	Auditability	Admin actions, score computations and payments shall be logged.
NFR-11	AI quality	Fair-price model target: median error within 15% on validation data (to be refined); AI shall degrade gracefully.
NFR-12	Localization	UI text shall be externalised to support Hindi and additional languages.

Note: numeric targets above are initial proposals to be validated during design and testing.

6. Use Cases

UC-01: Register and Verify Seller

Field	Description
Actor	Seller, Admin
Preconditions	User has a mobile number.
Main flow	1. User enters mobile number. 2. System sends OTP. 3. User verifies OTP. 4. User uploads ID proof. 5. Admin reviews and approves.
Alternate/exception flows	Invalid or expired OTP: system allows retry. ID rejected: user is told the reason and can resubmit.
Postconditions	Seller account is verified and can publish listings.

UC-02: Create Vehicle Listing

Field	Description
Actor	Seller
Preconditions	Seller is logged in and verified.
Main flow	1. Seller enters vehicle details. 2. Seller captures guided photos. 3. Seller uploads documents. 4. System computes Health Score. 5. Seller publishes for sale and/or rent.
Alternate/exception flows	Missing mandatory photo/document: system blocks publish and shows what is missing. Duplicate registration number: system flags listing.
Postconditions	Listing is live or pending verification.

UC-03: Search and View Vehicle

Field	Description
Actor	Buyer
Preconditions	None.
Main flow	1. Buyer searches or applies filters. 2. System shows ranked results with Health Score. 3. Buyer opens a listing and views details and fair-price indicator.
Alternate/exception flows	No results: system suggests relaxed filters.
Postconditions	Buyer can save the listing or contact the seller.

UC-04: Use Sell-or-Rent Calculator

Field	Description
Actor	Owner
Preconditions	Owner is logged in.
Main flow	1. Owner enters vehicle details and costs. 2. System estimates sale value and net rental income. 3. System calculates break-even and shows recommendation. 4. Owner creates a listing from the result.
Alternate/exception flows	Insufficient price data: system shows a range with a low-confidence note. Vehicle not rental-eligible: system shows a warning.
Postconditions	Owner receives a recommendation with assumptions.

UC-05: Book a Rental Vehicle

Field	Description
Actor	Renter, Owner
Preconditions	Renter is verified with a valid driving licence; vehicle is available.
Main flow	1. Renter selects dates and sends request. 2. Owner accepts. 3. Renter pays rent and deposit. 4. Pickup photos are recorded. 5. Booking is confirmed.
Alternate/exception flows	Owner declines or times out: request is cancelled and no charge is made. Payment failure: booking is not confirmed.
Postconditions	Booking is confirmed and the calendar is blocked.

UC-06: Return Vehicle and Settle Deposit

Field	Description
Actor	Renter, Owner, Admin
Preconditions	Booking is active.
Main flow	1. Return photos are recorded. 2. Owner confirms return or raises a claim. 3. System releases deposit or holds it for dispute. 4. Both sides review each other.
Alternate/exception flows	Damage claim: admin reviews photos and decides the deposit adjustment.
Postconditions	Booking closed; deposit settled.

UC-07: Get AI Recommendation

Field	Description
Actor	Buyer
Preconditions	None.
Main flow	1. Buyer describes needs. 2. System extracts constraints and ranks listings. 3. System shows top options with reasons.
Alternate/exception flows	AI service unavailable: system falls back to filter-based suggestions.
Postconditions	Buyer sees personalised suggestions.

UC-08: Moderate Listing and Resolve Report

Field	Description
Actor	Admin
Preconditions	Report or flag exists.
Main flow	1. Admin opens the review queue. 2. Admin checks evidence. 3. Admin approves, edits, removes or suspends. 4. System notifies the parties.
Alternate/exception flows	Insufficient evidence: admin requests more information.
Postconditions	Report closed with an audit entry.

7. Requirement Priorities and Traceability

Mapping of objectives to requirement groups:

Project objective	Related requirements	Release
One platform for buy/sell and rental	FR-LM, FR-SD, FR-RM	MVP
Reduce fraud and build trust	FR-UM-04/05, FR-HS, FR-CR-04, FR-AD, FR-AI-03	MVP and Beta
Fair price visibility	FR-AI-02, FR-SR-02	Beta and AI phase
Help owners choose to sell or rent	FR-SR	Beta
Personalised recommendations	FR-AI-01, FR-AI-04, FR-AI-05	AI phase
Secure transactions	FR-PM, FR-RM-05/07, NFR-04	MVP and Beta

Release split: Must-have items form the MVP; Should-have items arrive in Beta and AI phases; Could-have items are scheduled after validation.

8. Appendix

A. Sell-or-Rent Calculation Example (illustrative only)

Assume a bike with an estimated sale value of 60,000 rupees. Daily rate is 500 rupees, expected rented days per month is 12, so gross monthly rent is 6,000 rupees. Monthly costs (maintenance, insurance share, platform fee, cleaning) are 1,800 rupees, so net rental income is 4,200 rupees. Break-even is 60,000 / 4,200, which is about 14.3 months. If the bike is expected to lose around 8,000 rupees of value in that time, the owner should compare the net rental income against that depreciation, and also consider legal eligibility for renting. These numbers are examples, not market data.

B. Health Score Calculation Example

Documents 24/30, service and challan 12/20, photos 16/20, seller verification 15/15, reviews 9/15 gives a total of 76/100, which falls in the Good band.

C. Revision History

Version	Date	Author	Description
1.0	24 Oct 2026	Project Team	Initial draft of documentation and SRS
1.1	24 Oct 2026	Project Team	Frontend stack changed to React (Vite) + Tailwind CSS in JavaScript, matching team skills

D. Open Items

- Finalise legal position on rentals and the launch city/segment.
- Confirm available and authorised vehicle-verification data sources.
- Validate Health Score weights and pricing with user interviews.
- Select final technology stack and hosting.